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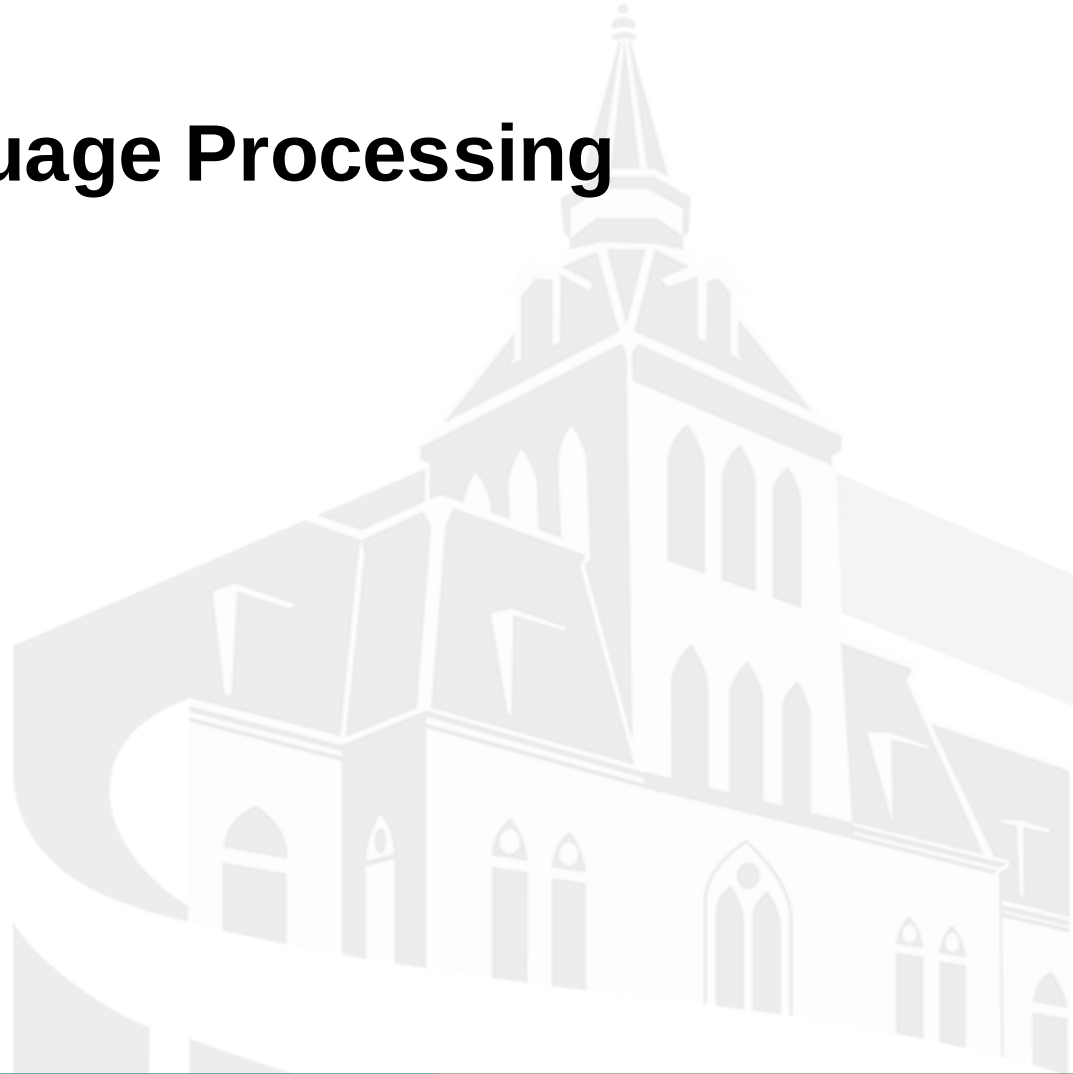


CS 584 Natural Language Processing

Language Modeling

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Learning Objectives

- Explain what Language Modeling (LM) is
- Implement N-gram LM
- Explain Perplexity
- Apply Good-turing and Kneser-Ney smoothing



Language modeling

- LANGUAGE is a prominent ability in **human beings** to express and communicate, which develops in early childhood and evolves over a lifetime.
- **Machines**, however, cannot naturally grasp the abilities of understanding and communicating in the form of human language, unless equipped with powerful artificial intelligence (AI) algorithms.
- It has been a longstanding research challenge to achieve this goal, to enable machines to **read, write, and communicate like humans**.

A Survey of Large Language Models: <https://arxiv.org/pdf/2303.18223.pdf>



Language modeling

- In general, language modeling is the task of **predicting what word comes next**.
- For example, “the students opened their [blank]” (books? laptops? minds? exams?)
- Formally, given a **sequence** of words $x^{(1)}, x^{(2)}, \dots, x^{(t)}$, compute the **probability distribution of the next word** $x^{(t+1)}$:
$$p(x^{(t+1)} = w_j | x^{(t)}, \dots, x^{(1)})$$

where w_j is a word in the vocabulary $V = \{w_1, \dots, w_{|V|}\}$.
- A system that does this is called a **language model**.

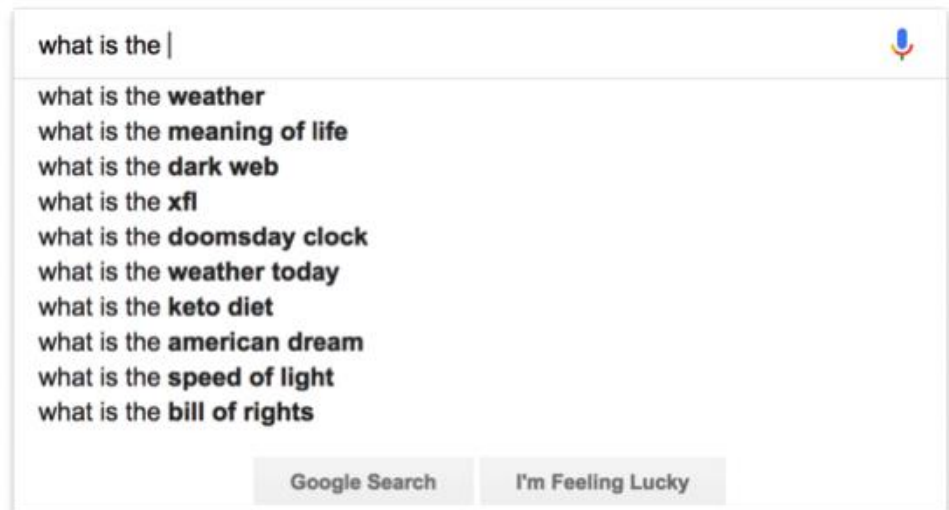
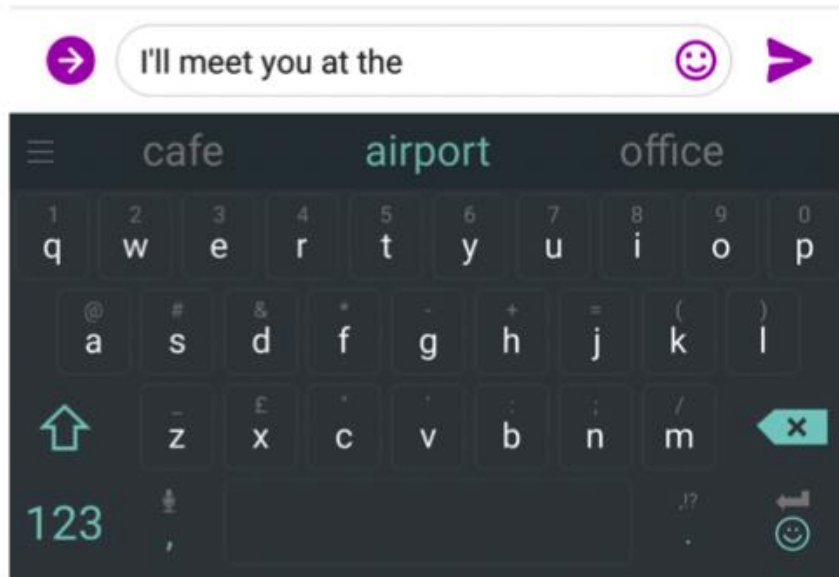


Why should we care about LM?

- LM is a benchmark task that helps us **measure our progress** on understanding language.
- LM is a **subcomponent** of many NLP tasks, especially those involving **generating text** or **estimating the probability of text**:
 - Predictive typing
 - Speech recognition
 - Handwriting recognition
 - Spelling/grammar correction
 - Authorship identification
 - Machine translation
 - Summarization
 - Dialogue



We use Language Models every day!





Development of LM 1: Statistical LM

- Developed based on **statistical** learning methods in 1990s.
- N-gram LMs, such as bigram and trigram.
- Limitations:
 - Difficult to accurately estimate high-order language models since an exponential number of transition probabilities need to be estimated.
 - Specially designed smoothing strategies are used to alleviate the data sparsity problem.



Development of LM 2: Neural LM

- Characterize the probability of word sequences by **neural networks**, e.g. RNN
- Introduce the concept of **distributed representation of words**, such as word2vec, which were demonstrated to be very effective across a variety of NLP tasks.
- These studies have **initiated the use of language models for representation learning**, having an important impact on the field of NLP.



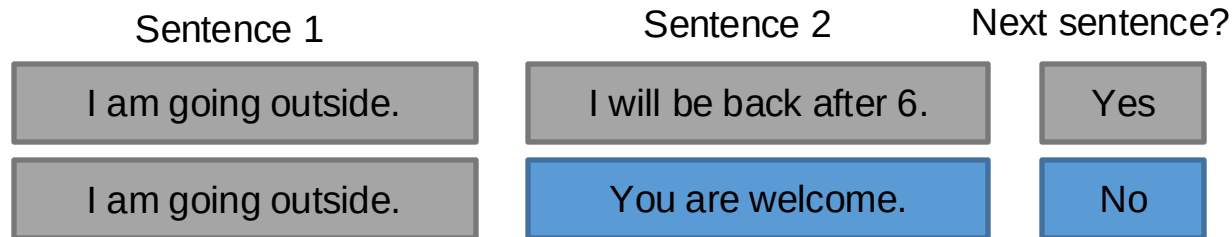
Development of LM 3: Pre-trained LM (PLM)

- **Pre-training first and then fine-tuning** on specific downstream task, allows to capture context-aware word representation.
- BERT: pretraining with specially designed tasks on large-scale **unlabeled** data, which is very effective to serve a **general-purpose** semantic features and improves the performance of NLP tasks.
- Inspired many follow-up work which sets the “**pre-training and fine-tuning**” **learning paradigm**: with different architectures or improved pre-training strategies.

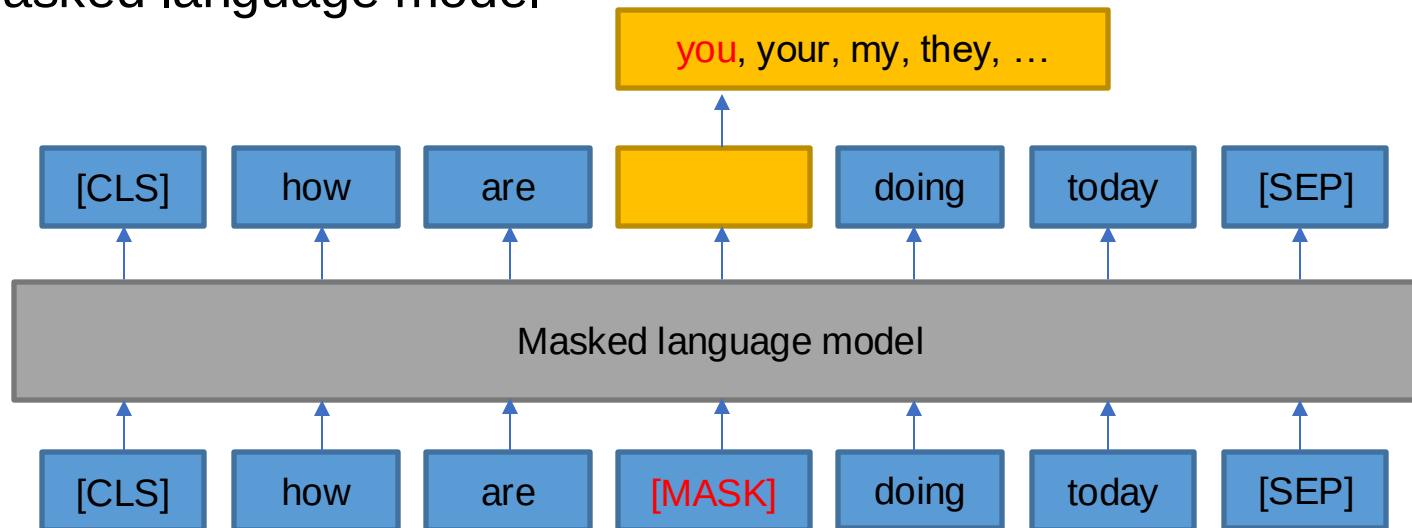


Pre-training BERT

- Next sentence prediction



- Masked language model



Devlin J, et al., Bert: Pre-training of deep bidirectional transformers for language understanding, 2018.



Development of LM 4: Large LM (LLM)

- **Scaling PLM in model size** often leads to an **emergent model capacity**.
- For example, GPT-3 can solve few-shot tasks through in-context learning, while GPT-2 cannot do well.
- The term LLM is coined for the **large-sized PLMs**.
- An application of LLMs is ChatGPT that adapts the LLMs from the GPT series for dialogue.
- Followed with a sharp release of LLMs: Bard, Claude 2, LLaMA, Falcon, etc.



Statistical Language Modeling



N-gram language models

- Question: how to learn a language model?
- Answer (pre-deep learning): learn an **n-gram** language model.
- Definition: An n-gram is a chunk of n consecutive words.
 - **Uni**-grams: “the”, “students”, “opened”, “their”
 - **Bi**-grams: “the students”, “students opened”, “opened their”
 - **Tri**-grams: “the students opened”, “students opened their”
 - **4**-grams: “the students opened their”
- Idea: Collect **statistics** about how frequent different n-grams are, and use these to predict next word.

N-gram language models

- We make a **simplifying assumption**: the next word only depends on the preceding $n-1$ words
- Thus, the probability can be written as:

$$P(x^{(t+1)} | x^{(t)}, \dots, x^{(t-n+2)}) = \frac{P(x^{(t+1)}, x^{(t)}, \dots, x^{(t-n+2)})}{P(x^{(t)}, \dots, x^{(t-n+2)})}$$

Prob. of a n -gram

Prob. of a $(n-1)$ -gram

- Q: How do we get these n -gram and $(n-1)$ -gram probabilities?



N-gram language models

- Statistical approximation: By counting them in large text corpus!

$$P(x^{(t+1)} | x^{(t)}, \dots, x^{(t-n+2)}) = \frac{P(x^{(t+1)}, x^{(t)}, \dots, x^{(t-n+2)})}{P(x^{(t)}, \dots, x^{(t-n+2)})}$$

- Assuming a 4-gram language model:

$$p(w_j | \text{"students opened their"}) = \frac{\text{count}(\text{students opened their } w_j)}{\text{count}(\text{students opened their})}$$

For example, suppose that in this corpus:

- “students opened their” occurred 1000 times
- “students opened their books” occurred 400 times
- -> $P(\text{"books"} | \text{"students opened their"}) = 0.4$
- “students opened their exams” occurred 100 times
- -> $P(\text{"exams"} | \text{"students opened their"}) = 0.1$



How to compute joint probability

- How to compute this joint probability:
 - $P(\text{its, water, is, so, transparent, that})$
- Intuition: let's rely on the **Chain Rule** of Probability



Reminder: the chain rule

- Recall the definition of conditional probability
- More variables:
 - $P(A,B,C,D) = P(A)P(B|A)P(C|A,B)P(D|A,B,C)$
- The Chain Rule in General
 - $P(x_1, x_2, x_3, \dots, x_n) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2) \dots P(x_n|x_1, \dots, x_{n-1})$



The chain rule for joint probability

$$P(w_1 w_2 \dots w_n) = \prod_i P(w_i | w_1 w_2 \dots w_{i-1})$$

$P(\text{"its water is so transparent"}) =$

$P(\text{its}) \times P(\text{water} | \text{its}) \times P(\text{is} | \text{its water})$

$\times P(\text{so} | \text{its water is}) \times P(\text{transparent} | \text{its water is so})$



How to estimate these probabilities

Can we just simply **count and divide**?

$$P(\text{"the"} \mid \text{"its water is so transparent that"}) = \frac{\text{count (its water is so transparent that the)}}{\text{count (its water is so transparent that)}}$$

No! Too many possible sentences!

We'll never see enough data for estimating these.



Markov Assumption

- The probability of moving to the next state **depends only on the present state** and not on the previous states.
- Simplifying assumption:

$$P(\text{"the"} \mid \text{"its water is so transparent that"}) = P(\text{"the"} \mid \text{"that"})$$

- Or maybe:

$$\begin{aligned} P(\text{"the"} \mid \text{"its water is so transparent that"}) \\ = P(\text{"the"} \mid \text{"transparent that"}) \end{aligned}$$



Simplest case: unigram model

$$P(w_1 \dots w_n) \approx \prod_i P(w_i)$$

- Some automatically generated sentences from a unigram model:
 - *fifth, an, of, futures, the, an, incorporated, a, a, the, inflation, most, dollars, quarter, in, is, mass*
 - *thrift, did, eighty, said, hard, 'm, july, bullish'*
 - *that, or, limited, the*



Bigram model

- Condition on the previous word:

$$P(w_1, \dots, w_n) \approx \prod_i P(w_i | w_{i-1})$$

- Some automatically generated sentences from a bigram model:
 - *texaco, rose, one, in, this, issue, is, pursuing, growth, in, a, boiler, house, said, mr., gurria, mexico, 's, motion, control, proposal, without, permission, from, five, hundred, fifty, five, yen*
 - *outside, new, car, parking, lot, of, the, agreement, reached*
 - *this, would, be, a, record, november*



N-gram model

- We can extend to trigrams, 4-grams, 5-grams
- In general, this is an insufficient model of language because language has **long-distance dependencies**:
 - *“The computer which I had just put into the machine room on the fifth floor crashed”*
- But we can often get away with N-gram models



Estimating probabilities

- The **maximum likelihood estimate (MLE)**
 - of some parameter of a model M from a training set T
 - maximizes the likelihood of the training set T given the model M
- Suppose the word “bagel” occurs 400 times in a corpus of a million words
- What is the probability that a random word from some other text will be “bagel”?
- MLE estimate is $400/1,000,000 = .0004$
- This may be a bad estimate for some other corpus
 - But it is the estimate that makes it most likely that “bagel” will occur 400 times in a million word corpus.

Estimating bigram probabilities

- The Maximum Likelihood Estimate
- Bigram probability of a word w_i given a previous word w_{i-1} :

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}w_i)}{\sum_w c(w_{i-1}w)}$$



$$P(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

An example

- $\langle s \rangle$ I am Sam $\langle /s \rangle$
- $\langle s \rangle$ Sam I am $\langle /s \rangle$
- $\langle s \rangle$ I do not like green eggs and ham $\langle /s \rangle$

$$P(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

$$P(I \mid \langle s \rangle) = \frac{2}{3} = 0.67$$

$$P(\text{Sam} \mid \langle s \rangle) = \frac{1}{3} = 0.33$$

$$P(\text{am} \mid I) = \frac{2}{3} = 0.67$$

$$P(\langle /s \rangle \mid \text{Sam}) = \frac{1}{2} = 0.5$$

$$P(\text{Sam} \mid \text{am}) = \frac{1}{2} = 0.5$$

$$P(\text{do} \mid I) = \frac{1}{3} = 0.33$$



Practical Issues

- For computing efficiency, we do everything in **log space**
 - Avoid underflow: the probabilities become so close to zero that they cannot be accurately represented.
 - Also adding is faster than multiplying

$$\log (p_1 \times p_2 \times p_3 \times p_4) = \log p_1 + \log p_2 + \log p_3 + \log p_4$$



Language Modeling Toolkits

- SRILM - The SRI Language Modeling Toolkit
 - <http://www.speech.sri.com/projects/srilm/>
- The CMU-Cambridge Statistical Language Modeling Toolkit v2
 - http://www.cs.cmu.edu/~dorcass/toolkit_documentation.html



Google N-gram release (2006)

- [All Our N-gram are belong to you](#)

“We processed 1,024,908,267,229 words of running text and are publishing the counts for all 1,176,470,663 five-word sequences that appear at least 40 times. There are 13,588,391 unique words, after discarding words that appear less than 200 times.”

- [Google N-gram viewer](#): A tool that displays a graph showing how those phrases have occurred in a corpus of books.



Sparsity problems in N-grams

Problem: what if “its water is so transparent that w” never occurred in data? Then w has probability 0!

(Partial) solution: add small value to the count for every w in the vocabulary. This is called **smoothing**.

$P(w \mid \text{“its water is so transparent that”}) =$

$\frac{\text{count (its water is so transparent that w)}}{\text{count (its water is so transparent that)}}$

Problem: what if “its water is so transparent that” never occurred in data? Then we cannot calculate probability for any w.

(Partial) solution: Just condition on “so transparent that” instead. This is called **backoff**.

Note: increasing n makes sparsity problems worse. Typically, we cannot have n bigger than 5.



Storage problems with n-gram

Storage: Need to store count for all n-grams you saw in the corpus

$P(w \mid \text{"its water is so transparent that"}) =$

count (its water is so transparent that w)

count (its water is so transparent that)

Note: increasing n or increasing corpus makes model size bigger!



Statistical Language Modeling

Goal: assign a probability to a sentence

❑ Machine Translation:

❑ $P(\text{high winds tonite}), P(\text{large winds tonite})$

❑ Spell Correction

❑ The office is about fifteen minuets from my house

❑ $P(\text{about fifteen minutes from}), P(\text{about fifteen minuets from})$

❑ Speech Recognition

❑ $P(\text{I saw a van}), P(\text{eyes awe of an})$

❑ + Summarization, question-answering, etc., etc.!



Generating text with an n-gram LM

You can use a language model to generate text:

today the price of gold per ton , while production of shoe lasts and shoe industry , the bank intervened just after it considered and rejected an imf demand to rebuild depleted european stocks , sept 30 end primary 76 cts a share.

Surprisingly grammatical!

But **incoherent**. We need to consider more than three words at a time if we want to model language well.



Extrinsic evaluation of N-gram models

- Does our language model prefer good sentences to bad ones?
 - Assign higher probability to “real” or “frequently observed” sentences than “ungrammatical” or “rarely observed” sentences?
- Best evaluation for comparing models A and B: extrinsic evaluation
 - Put each model in a **downstream task**
 - spelling corrector, speech recognizer, machine translation
 - Run the task, get an accuracy for A and for B
 - How many misspelled words corrected properly
 - How many words translated correctly
 - Compare accuracy for A and B
 - Limitation: Time-consuming; can take days or weeks



Intrinsic evaluation of N-gram models

- Sometimes use intrinsic evaluation: **perplexity**
 - Bad approximation
 - Unless the test data looks just like the training data
 - So generally, only useful in pilot experiments
 - But it is helpful to think about.



Perplexity

- The best language model is one that best predicts an unseen test set
 - Gives the highest $P(\text{sentence})$

$$P(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}$$

Chain rule:

$$= \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_1 \dots w_{i-1})}}$$

For bigrams:

$$P(W) = \sqrt[N]{\prod_{i=1}^N \frac{1}{P(w_i | w_{i-1})}}$$

- Perplexity is the inverse probability of the test set, normalized by the number of words
 - minimizing perplexity is the same as maximizing probability



Perplexity as a branching factor

- Let's suppose a sentence '0,1,2,3,4,5,6,7,8,9' consisting of random digits. How hard is the task of recognizing digits?
- Or what is the perplexity of this sentence according to a model that assign $P = 1/10$ to each digit?

$$\begin{aligned}P(W) &= P(w_1 w_2 \dots w_N)^{-\frac{1}{N}} \\&= \left(\frac{1}{10}\right)^N)^{-\frac{1}{N}} \\&= 10^{-1} \\&= 10\end{aligned}$$

- Perplexity 10 since each digit is equally likely.



Low perplexity = better model

- Training 38 millions words, test 1.5 million words (WSJ)

N-gram Order	Unigram	Bigram	Trigram
Perplexity	962	170	109



Overfitting

- N-grams only work well for word prediction if the test corpus looks like the training corpus
 - In real life, it often doesn't
 - We need to train robust models that **generalize!**
- One kind of generalization: **Zeros!**
 - Things that don't ever occur in the training set
 - But occur in the test set



Zeros: Zero probability bigrams

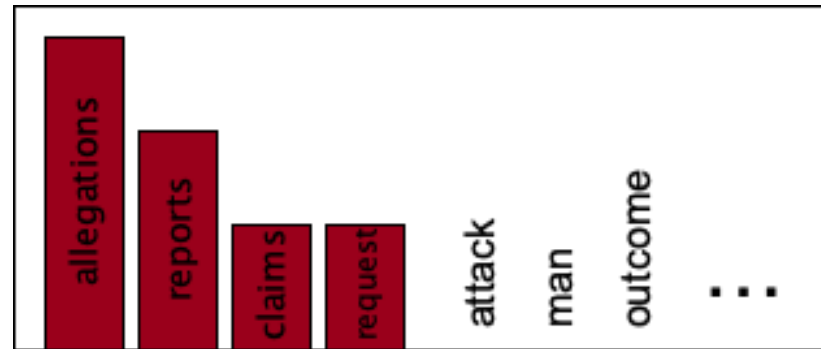
- Training set:
 - ... denied the allegations
 - ... denied the reports
 - ... denied the claims
 - ... denied the request
- Test set:
 - ... denied the offer
 - ... denied the loan
- Bigram with zero probability: means that we will assign 0 probability to the test set
- Hence, we cannot compute perplexity (divided by 0!)
- Solution: **Laplace smoothing**

$P(\text{"offer"} \mid \text{denied the}) = 0$
Perplexity = ∞

The intuition of smoothing

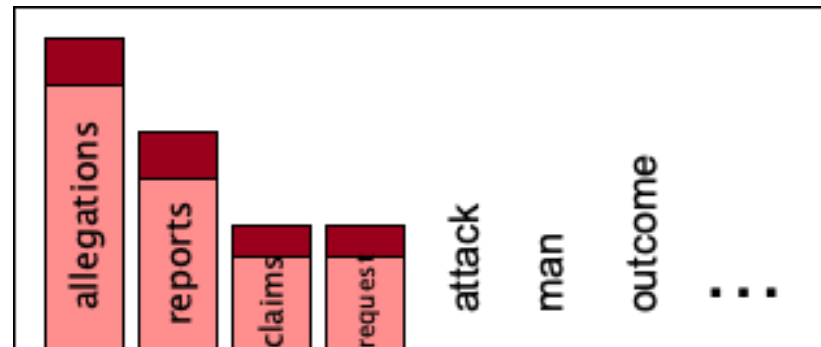
- When we have sparse statistics

- $P(w \mid \text{"denied the"})$
- 3 allegations
- 2 reports
- 1 claims
- 1 request
- 7 total



- Steal probability mass to generalize better

- $P(w \mid \text{"denied the"})$
- 2.5 allegations
- 1.5 reports
- 0.5 claims
- 0.5 request
- 2 others
- 7 total





Laplace smoothing

- Also called **add-one estimation**
- Pretend we saw each word **one more time** than we did
- Just add one to all the counts
- MLE estimate:

$$P_{\text{MLE}}(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

- Add-1 estimate:

$$P_{\text{Add-1}}(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i) + 1}{c(w_{i-1}) + V}$$



Berkeley Restaurant Corpus: Raw bigram counts vs. Laplace smoothed bigram counts

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

	i	want	to	eat	chinese	food	lunch	spend
i	3.8	527	0.64	6.4	0.64	0.64	0.64	1.9
want	1.2	0.39	238	0.78	2.7	2.7	2.3	0.78
to	1.9	0.63	3.1	430	1.9	0.63	4.4	133
eat	0.34	0.34	1	0.34	5.8	1	15	0.34
chinese	0.2	0.098	0.098	0.098	0.098	8.2	0.2	0.098
food	6.9	0.43	6.9	0.43	0.86	2.2	0.43	0.43
lunch	0.57	0.19	0.19	0.19	0.19	0.38	0.19	0.19
spend	0.32	0.16	0.32	0.16	0.16	0.16	0.16	0.16



Add-1 estimation is a blunt instrument

- Primary limitations of Add-1 estimation:
 - **Overestimation of probabilities**, especially for less common n-grams
 - **Loss of discrimination**: It tends to smooth out differences between n-grams, leading to a loss of discrimination between more and less likely sequences.
 - **Sensitivity to vocabulary size**: The level of smoothing in "add-1" depends on the size of the vocabulary and the length of the n-gram. In cases where the vocabulary is large or the n-gram length is high, the added "pseudocount" of 1 may have a smaller impact on the estimated probabilities.
- So add-1 isn't used for N-grams
 - But add-1 is used to smooth other NLP models in domains where the number of zeros isn't so huge



Backoff and interpolation

- Backoff: Sometimes it helps to use **less context**
 - Condition on less context for contexts you haven't learned much about
 - Use trigram if you have sufficient evidence
 - Otherwise bigram, otherwise unigram
- Interpolation: mixing the probability estimates from all n-gram estimators; **weighting and combining** the unigram, bigram, and trigram counts works better.

Linear Interpolation

- Simple interpolation

$$\begin{aligned}\hat{P}(w_n|w_{n-1}w_{n-2}) &= \lambda_1 P(w_n|w_{n-1}w_{n-2}) \\ &\quad + \lambda_2 P(w_n|w_{n-1}) \\ &\quad + \lambda_3 P(w_n) \\ \sum_i \lambda_i &= 1\end{aligned}$$

- Lambdas conditional on context:

$$\begin{aligned}\hat{P}(w_n|w_{n-2}w_{n-1}) &= \lambda_1(w_{n-2}^{n-1}) P(w_n|w_{n-2}w_{n-1}) \\ &\quad + \lambda_2(w_{n-2}^{n-1}) P(w_n|w_{n-1}) \\ &\quad + \lambda_3(w_{n-2}^{n-1}) P(w_n)\end{aligned}$$

Now lambdas are dependent on what the previous two words were.



How to set the lambdas?

- Use a held-out corpus



- Choose lambdas to maximize the probability of held-out data:
 - fix the N-gram probabilities (on the training data)
 - then search for lambdas that give largest probability to held-out set.



Unknown words:

open versus closed vocabulary task

- If we know all the words in advanced
 - Vocabulary V is fixed
 - Closed vocabulary task
- Often we don't know this
 - Out Of Vocabulary = OOV words
 - Open vocabulary task
- Instead: create an unknown word token $\langle \text{UNK} \rangle$
 - Training of $\langle \text{UNK} \rangle$ probabilities
 - Create a fixed lexicon L of size V
 - At text normalization phase, any training word not in L changed to $\langle \text{UNK} \rangle$
 - Now we train its probabilities like a normal word
 - At decoding time
 - If text input: Use UNK probabilities for any word not in training



Huge web-scale n-grams

- How to deal with, e.g., Google N-gram corpus
- Pruning
 - Only store N-grams with count $>$ threshold.
 - Remove singletons of higher-order n-grams
 - Entropy-based pruning
- Efficiency
 - Efficient data structures like tries
 - Bloom filters: approximate language models
 - Store words as indexes, not strings
 - Use Huffman coding to fit large numbers of words into two bytes
 - Quantize probabilities (4-8 bits instead of 8-byte float)



Smoothing for web-scale N-grams

- “Stupid backoff” (Brants *et al.* 2007)
- No discounting, just use relative frequencies

$$S(w_i | w_{i-k+1}^{i-1}) = \begin{cases} \frac{\text{count}(w_{i-k+1}^i)}{\text{count}(w_{i-k+1}^{i-1})} & \text{if } \text{count}(w_{i-k+1}^i) > 0 \\ 0.4S(w_i | w_{i-k+2}^{i-1}) & \text{otherwise} \end{cases}$$

The recursion ends at unigrams

$$S(w_i) = \frac{\text{count}(w_i)}{N}$$



N-gram smoothing summary

- Add-1 smoothing:
 - OK for text categorization, not for language modeling
- The most commonly used method:
 - Extended Interpolated Kneser-Ney
- For very large N-grams like the Web:
 - Stupid backoff



Advanced language modeling

- Discriminative models:
 - choose n-gram weights to improve a task, not to fit the training set

- Parsing-based models

- Caching Models

- Recently used words are more likely to appear

$$P_{CACHE}(w | history) = \lambda P(w_i | w_{i-2} w_{i-1}) + (1 - \lambda) \frac{c(w \in history)}{|history|}$$

- These perform very poorly for speech recognition (why?)



Reminder: add-1 (laplace) smoothing

$$P_{\text{Add-1}}(w_i | w_{i-1}) = \frac{c(w_{i-1}, w_i) + 1}{c(w_{i-1}) + V}$$



More general formulations: add-k

$$P_{add-k}(w_i|w_{i-1}) = \frac{c(w_{i-1}, w_i) + k}{c(w_{i-1}) + kV}$$

$$P_{add-k}(w_i|w_{i-1}) = \frac{c(w_{i-1}, w_i) + m(\frac{1}{V})}{c(w_{i-1}) + m}$$



Unigram prior smoothing

$$P_{add-k}(w_i|w_{i-1}) = \frac{c(w_{i-1}, w_i) + m(\frac{1}{V})}{c(w_{i-1}) + m}$$

$$P_{\text{unigramprior}}(w_i|w_{i-1}) = \frac{c(w_{i-1}, w_i) + mP(w_i)}{c(w_{i-1}) + m}$$



Advanced smoothing algorithms

- Intuition used by many smoothing algorithms
 - Good-Turing
 - Kneser-Ney
 - Witten-Bell
- Use the count of things we've seen once
 - to help estimate the count of things we've never seen



Frequency of frequency c

- N_c = the count of things we've seen c times
 - Emily I am I am Emily I do not eat

I 3

Emily 2

am 2

do 1

not 1

eat 1

$$N_1 = 3$$

$$N_2 = 2$$

$$N_3 = 1$$



Good Turing estimation

- Idea: reallocate the probability mass of n-grams that **occur c+1 times** in the training data to the n-grams that **occur c times**
- In particular, reallocate the probability mass of n-grams that were **seen once** to the n-grams that were **never seen**,
- For each count c, we compute an adjusted count c^* :

$$c^* = \frac{(c+1)N_{c+1}}{N_c}$$

where N_c is the number of n-grams seen exactly c times.

- Then we have:

$$P_{GT}^* = c^* / N$$

where
$$N = \sum_{r=1}^{\infty} cN_c$$



Good Turing smoothing intuition

- You are fishing and caught:
 - 10 carps, 3 perch, 2 whitefish, 1 trout, 1 salmon, 1 eel
= 18 fish
- How likely is it that the next species is trout?
 - 1/18
- How likely is it that the next species is new (i.e. catfish or bass)?
 - use our estimate of things-we-saw-once to estimate the news things: $3/18$ ($N_1=3$)
- Assuming so, how likely is it that the next species is trout?
 - must be less than 1/18, how to estimate?



Good Turing calculations

- P_{GT}^* (things with zero frequency) = N_1/N
- Other frequencies: $c^* = \frac{(c+1)N_{c+1}}{N_c}$

- Unseen (bass or catfish)

- $c=0$
- $MLE_p = 0/18 = 0$
- $P_{GT}^*(\text{unseen}) = N_1/N = 3/18$

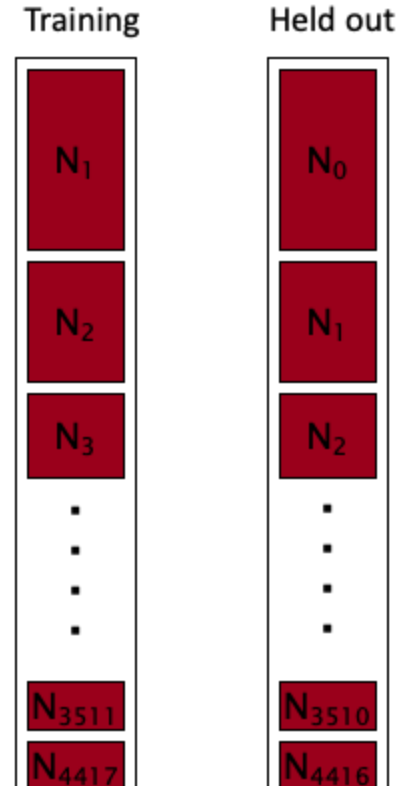
- Seen once (trout)

- $c = 1$,
- $MLE_p = 1/18$
- $c^*(\text{trout}) = 2 * N_2/N_1 = 2 * 1/3 = 2/3$
- $P_{GT}^*(\text{trout}) = 2/3/18 = 1/27$

Ney et al. Good Turing Intuition

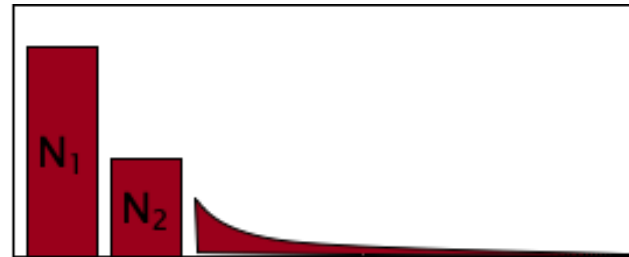
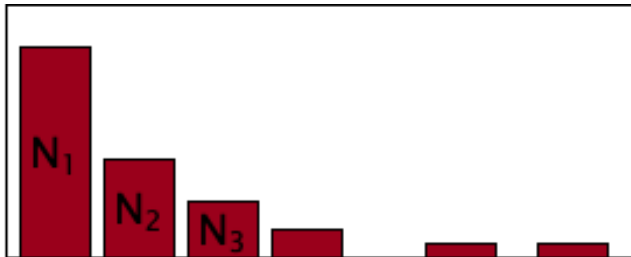
- Intuition from leave-one-out validation

- Take each of the c training words out in turn
- c training sets of size $c-1$, held-out of size 1
- What fraction of held-out words are unseen in training?
 - N_1/c
- What fraction of held-out words are seen k times in training?
 - $(k+1)N_{k+1}/c$
- So in the future we expect $(k+1)N_{k+1}/c$ of the words to be those with training count k
- There are N_k words with training count k
- Each should occur with probability:
 - $(k+1)N_{k+1}/c/N_k$
- ...or expected count: $k^* = \frac{(k+1)N_{k+1}}{N_k}$



Good-Turing complications

- Problem: what about “the”? (say $c=4417$)
 - For small k , $N_k > N_{k+1}$
 - For large k , too jumpy, zeros wreck estimates
 - Simple Good-Turing [Gale and Sampson]: replace empirical N_k with a best-fit power law once counts get unreliable





Resulting Good-Turing numbers

- Numbers from Church and Gale(1991)
- 22 million words of AP Newswire

$$c^* = \frac{(c+1)N_{c+1}}{N_c}$$

- It sure looks like $c^* = (c - .75)$

Count c	Good Turing c*
0	.0000270
1	0.446
2	1.26
3	2.24
4	3.24
5	4.22
6	5.19
7	6.21
8	7.24
9	8.25



Absolute Discounting Interpolation

- Same ourselves some time and just subtract 0.75 (or some d)!

$$P_{\text{AbsoluteDiscounting}}(w_i | w_{i-1}) = \frac{\overset{\text{discounted bigram}}{c(w_{i-1}, w_i) - d}}{c(w_{i-1})} + \overset{\text{Interpolation weight}}{\lambda(\overset{\swarrow}{w_{i-1}})} \underset{\nwarrow \text{unigram}}{P(w)}$$

- Maybe keeping a couple extra values of d from counts 1 and 2 (last slide)
- But should we really just use the regular unigram $P(w)$?



Kneser-Ney Smoothing I

- Better estimate for probabilities of lower-order unigrams!
 - Shannon game: I can't see without my reading_____?
 - “Francisco” is more common than “glasses”
 - ... but “Francisco” always follows “San”
- The unigram is useful exactly when we haven't seen this bigram!
- Instead of $P(w)$: “How likely is w ”
- $P_{\text{continuation}}(w)$: “How likely is w to appear as a novel continuation?”
 - For each word, count the number of bigram types it completes
 - Every bigram type was a novel continuation the first time it was seen:

$$P_{\text{CONTINUATION}}(w) \propto |\{w_{i-1} : c(w_{i-1}, w) > 0\}|$$



Kneser-Ney Smoothing II

- How many times does w appear as a novel continuation:

$$P_{CONTINUATION}(w) \propto |\{w_{i-1} : c(w_{i-1}, w) > 0\}|$$

- Normalized by the total number of word bigram types

$$|\{(w_{j-1}, w_j) : c(w_{j-1}, w_j) > 0\}|$$

$$P_{CONTINUATION}(w) = \frac{|\{w_{i-1} : c(w_{i-1}, w) > 0\}|}{|\{(w_{j-1}, w_j) : c(w_{j-1}, w_j) > 0\}|}$$



Kneser-Ney Smoothing III

- Alternative metaphor: The number (#) of word types seen to precede w

$$|\{w_{i-1} : c(w_{i-1}, w) > 0\}|$$

- Normalized by the # of words preceding all words:

$$P_{CONTINUATION}(w) = \frac{|\{w_{i-1} : c(w_{i-1}, w) > 0\}|}{\sum_{w'} |\{w'_{i-1} : c(w'_{i-1}, w') > 0\}|}$$

- A frequent word (Francisco) occurring in only one context (San) will have a low continuation probability



Kneser-Ney Smoothing IV

$$P_{KN}(w_i | w_{i-1}) = \frac{\max(c(w_{i-1}, w_i) - d, 0)}{c(w_{i-1})} + \lambda(w_{i-1})P_{CONTINUATION}(w_i)$$

- λ is a normalizing constant; the probability mass we've discounted

$$\lambda(w_{i-1}) = \frac{d}{c(w_{i-1})} |\{w : c(w_{i-1}, w) > 0\}|$$

the normalized discount

The number of word types that can follow w_{i-1}
= # of word types we discounted
= # of times we applied normalized discount



Kneser-Ney Smoothing Recursive formulation

$$P_{KN}(w_i | w_{i-n+1}^{i-1}) = \frac{\max(c_{KN}(w_{i-n+1}^i) - d, 0)}{c_{KN}(w_{i-n+1}^{i-1})} + \lambda(w_{i-n+1}^{i-1}) P_{KN}(w_i | w_{i-n+2}^{i-1})$$

$$c_{KN}(\bullet) = \begin{cases} \text{count}(\bullet) & \text{for the highest order} \\ \text{continuationcount}(\bullet) & \text{for lower order} \end{cases}$$

Continuation count = Number of unique single word contexts for ($\bullet$)



Readings

- CH 9, 13 SLP; CH 14 NNLP
- A Survey of Large Language Models: <https://arxiv.org/pdf/2303.18223.pdf>
- N-gram language models: <https://web.stanford.edu/~jurafsky/slp3/3.pdf>



Acknowledgements

Slides adapted from Dr. Dan Jurafsky's Introduction to Natural Language Processing at Stanford.



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Thank You